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LIGHT EMITTING DEVICE FOR DISPLAY HAVING LED STACKS

Abstract

A light emitting device for a display including a circuit board, a plurality of light emitting units arranged on the circuit board, each light emitting unit comprising a first LED stack including a first conductivity type semiconductor layer, a second conductivity type semiconductor layer, a first electrode disposed on the first conductivity type semiconductor layer, and a second electrode disposed on the second conductivity type semiconductor layer, a plurality of bump pads disposed between the plurality of light emitting units and the circuit board, and a bonding layer disposed between the second electrode and the circuit board, in which the second electrode has a side surface recessed inwardly with respect to a side surface of the light emitting unit to define a recessed portion, and the bonding layer is filled in the recessed portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation of U.S. application Ser. No. 17/826,106, filed May 26, 2022, which is a Continuation of U.S. application Ser. No. 16/782,594, filed Feb. 5, 2020 (now U.S. Pat. No. 11,362,073), which claims priority from and the benefit of U.S. Provisional Application No. 62/802,784, filed on Feb. 8, 2019, each of which is hereby incorporated by reference for all purposes as if fully set forth herein.

BACKGROUND

Field

[0002] Exemplary embodiments relate to a light emitting device for a display and a display apparatus, and, more particularly, to a light emitting device having a stack structure of a plurality of LEDs for a display, and a display apparatus including the same.

Discussion of the Background

[0003] As an inorganic light source, light emitting diodes have been used in various technical fields, such as displays, vehicular lamps, general lighting, and the like. With various advantages of the light emitting diodes, such as longer lifespan, lower power consumption, and faster response, than existing light sources, light emitting diodes have been replacing the existing light sources.

[0004] Light emitting diodes have been generally used as backlight light sources in display apparatuses. However, LED displays that directly realize images using the light emitting diodes have been recently developed.

[0005] In general, a display apparatus displays various colors through mixture of blue, green, and red light. In order to realize various images, the display apparatus includes a plurality of pixels, each including sub-pixels corresponding to one of blue, green, and red light. As such, a color of a certain pixel is typically determined based on the colors of the sub-pixels, so that images can be realized through the combination of such pixels.

[0006] Since LEDs can emit various colors depending upon materials thereof, individual LED chips emitting blue, green and red light may be arranged on a two-dimensional plane to provide a display apparatus. However, when one LED chip is provided to each sub-pixel, the number of LED chips may be increased, which may require excessive time for a mounting process during manufacture.

[0007] Moreover, since the sub-pixels are arranged on the two-dimensional plane in the display apparatus, a relatively large area is occupied by one pixel that includes the sub-pixels for blue, green, and red light. In this case, when an area of each sub-pixel is reduced to arrange the sub-pixels in a restricted area, luminous area of the sub-pixels may be reduced, thereby deteriorating the brightness of the pixels.

[0008] The above information disclosed in this Background section is only for understanding of the background of the inventive concepts, and, therefore, it may contain information that does not constitute prior art.

SUMMARY

[0009] Exemplary embodiments provide a light emitting device for a display that is capable of increasing an area of each sub-pixel in a restricted pixel area and a display apparatus including the same.

[0010] Exemplary embodiments also provide a light emitting device for a display that is capable of reducing a time associated with a mounting process and a display apparatus including the same.

[0011] Exemplary embodiments further provide a light emitting device for a display that is capable of increasing the production yield and a display apparatus including the same.

[0012] Additional features of the inventive concepts will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the inventive concepts.

[0013] A light emitting device for a display according to an exemplary embodiment includes: a first LED stack; a second LED stack disposed under the first LED stack; a third LED stack disposed under the second LED stack; a first transparent electrode interposed between the first LED stack and the second LED stack, and being in ohmic contact with a lower surface of the first LED stack; a second transparent electrode interposed between the first LED stack and the second LED stack, and being in ohmic contact with an upper surface of the second LED stack; a third transparent electrode interposed between the second LED stack and the third LED stack, and being in ohmic contact with an upper surface of the third LED stack; a first electrode pad disposed on a first conductivity type semiconductor layer of the third LED stack; a lower second electrode pad disposed on the third transparent electrode; and bump pads disposed on the first LED stack, in which each of the first to third LED stacks includes a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer, the bump pads include first, second, and third bump pads and a common bump pad, in which the common bump pad is commonly electrically connected to the first to third LED stacks, respectively, the first to third bump pads are electrically connected to the first to third LED stacks, respectively, and an upper surface of the first electrode pad is located at substantially the same elevation as that of the lower second electrode pad.

[0014] A light emitting device for a display according to another exemplary embodiment includes: a first LED stack; a second LED stack disposed under the first LED stack; a third LED stack disposed under the second LED stack; a first transparent electrode interposed between the first LED stack and the second LED stack, and being in ohmic contact with a lower surface of the first LED stack; a second transparent electrode interposed between the first LED stack and the second LED stack, and being in ohmic contact with an upper surface of the second LED stack; a third transparent electrode interposed between the second LED stack and the third LED stack, and being in ohmic contact with an upper surface of the third LED stack; bump pads disposed on the first LED stack, in which each of the first to third LED stacks includes a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer, and at least one of the first to third transparent electrodes is recessed from an edge of the second conductivity type semiconductor layer of the first to third LED stacks.

[0015] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the invention as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The accompanying drawings, which are included to provide a further understanding of the inventive concept, and are incorporated in and constitute a part of this specification, illustrate

exemplary embodiments of the inventive concept, and, together with the description, serve to explain principles of the inventive concepts.

[0017] FIG. 1 shows schematic perspective views illustrating display apparatuses according to exemplary embodiments.

[0018] FIG. 2 is a schematic plan view illustrating a display panel according to an exemplary embodiment.

[0019] FIG. 3A is a schematic plan view illustrating a light emitting device according to an exemplary embodiment.

[0020] FIG. 3B, FIG. 3C, and 3D are schematic cross-sectional views taken along the lines A-A', B-B', and C-C' of FIG. 3A, respectively.

[0021] FIG. 4A, FIG. 4B, and FIG. 4C are schematic cross-sectional views illustrating first, second, and third LED stacks grown on a growth substrate, respectively, according to an exemplary embodiment.

[0022] FIGS. 5A, 5B, 5C, 5D, 6A, 6B, 6C, 6D, 7A, 7B, 7C, 7D, 8A, 8B, 8C, 8D, 9A, 9B, 9C, 9D, 10A, 10B, 10C, 10D, 11A, 11B, 11C, 11D, 12A, 12B, 12C, 12D, 13A, 13B, 13C and 13D are schematic plan views and cross-sectional views illustrating a method of manufacturing a light emitting device for a display according to an exemplary embodiment.

[0023] FIG. 14 is a schematic cross-sectional view illustrating a light emitting device mounted on a circuit board according to an exemplary embodiment.

[0024] FIG. 15A, FIG. 15B, and FIG. 15C are schematic cross-sectional views illustrating a method of transferring a light emitting device to a circuit board according to an exemplary embodiment.

DETAILED DESCRIPTION

[0025] In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of various exemplary embodiments or implementations of the invention. As used herein “embodiments” and “implementations” are interchangeable words that are non-limiting examples of devices or methods employing one or more of the inventive concepts disclosed herein. It is apparent, however, that various exemplary embodiments may be practiced without these specific details or with one or more equivalent arrangements. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring various exemplary embodiments. Further, various exemplary embodiments may be different, but do not have to be exclusive. For example, specific shapes, configurations, and characteristics of an exemplary embodiment may be used or implemented in another exemplary embodiment without departing from the inventive concepts.

[0026] Unless otherwise specified, the illustrated exemplary embodiments are to be understood as providing exemplary features of varying detail of some ways in which the inventive concepts may be implemented in practice. Therefore, unless otherwise specified, the features, components, modules, layers, films, panels, regions, and/or aspects, etc. (hereinafter individually or collectively referred to as “elements”), of the various embodiments may be otherwise combined, separated, interchanged, and/or rearranged without departing from the inventive concepts.

[0027] The use of cross-hatching and/or shading in the accompanying drawings is generally provided to clarify boundaries between adjacent elements. As such, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, dimensions, proportions, commonalities between illustrated elements, and/or any other characteristic, attribute, property, etc., of the elements, unless specified. Further, in the accompanying drawings, the size and relative sizes of elements may be exaggerated for clarity and/or descriptive purposes. When an exemplary embodiment may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order. Also, like reference numerals

denote like elements.

[0028] When an element, such as a layer, is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected to, or coupled to the other element or layer or intervening elements or layers may be present. When, however, an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. To this end, the term “connected” may refer to physical, electrical, and/or fluid connection, with or without intervening elements. Further, the D1-axis, the D2-axis, and the D3-axis are not limited to three axes of a rectangular coordinate system, such as the x, y, and z—axes, and may be interpreted in a broader sense. For example, the D1-axis, the D2-axis, and the D3-axis may be perpendicular to one another, or may represent different directions that are not perpendicular to one another. For the purposes of this disclosure, “at least one of X, Y, and Z” and “at least one selected from the group consisting of X, Y, and Z” may be construed as X only, Y only, Z only, or any combination of two or more of X, Y, and Z, such as, for instance, XYZ, XYY, YZ, and ZZ. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0029] Although the terms “first,” “second,” etc. may be used herein to describe various types of elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the teachings of the disclosure.

[0030] Spatially relative terms, such as “beneath,” “below,” “under,” “lower,” “above,” “upper,” “over,” “higher,” “side” (e.g., as in “sidewall”), and the like, may be used herein for descriptive purposes, and, thereby, to describe one elements relationship to another element(s) as illustrated in the drawings. Spatially relative terms are intended to encompass different orientations of an apparatus in use, operation, and/or manufacture in addition to the orientation depicted in the drawings. For example, if the apparatus in the drawings is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. Furthermore, the apparatus may be otherwise oriented (e.g., rotated 90 degrees or at other orientations), and, as such, the spatially relative descriptors used herein interpreted accordingly.

[0031] The terminology used herein is for the purpose of describing particular embodiments and is not intended to be limiting. As used herein, the singular forms, “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Moreover, the terms “comprises,” “comprising,” “includes,” and/or “including,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. It is also noted that, as used herein, the terms “substantially,” “about,” and other similar terms, are used as terms of approximation and not as terms of degree, and, as such, are utilized to account for inherent deviations in measured, calculated, and/or provided values that would be recognized by one of ordinary skill in the art.

[0032] Various exemplary embodiments are described herein with reference to sectional and/or exploded illustrations that are schematic illustrations of idealized exemplary embodiments and/or intermediate structures. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, exemplary embodiments disclosed herein should not necessarily be construed as limited to the particular illustrated shapes of regions, but are to include deviations in shapes that result from, for instance, manufacturing. In this manner, regions illustrated in the drawings may be schematic in nature and the shapes of these regions may not reflect actual shapes of regions of a device and, as such, are not necessarily intended to be limiting.

[0033] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this

disclosure is a part. Terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized or overly formal sense, unless expressly so defined herein.

[0034] A light emitting device for a display according to an exemplary embodiment includes: a first LED stack; a second LED stack disposed under the first LED stack; a third LED stack disposed under the second LED stack; a first transparent electrode interposed between the first LED stack and the second LED stack, and being in ohmic contact with a lower surface of the first LED stack; a second transparent electrode interposed between the first LED stack and the second LED stack, and being in ohmic contact with an upper surface of the second LED stack; a third transparent electrode interposed between the second LED stack and the third LED stack, and being in ohmic contact with an upper surface of the third LED stack; an n-electrode pad disposed on a first conductivity type semiconductor layer of the third LED stack; a lower p-electrode pad disposed on the third transparent electrode; and bump pads disposed on the first LED stack, in which each of the first to third LED stacks includes a first conductivity type semiconductor layer, an active layer and a second conductivity type semiconductor layer, the bump pads include first to third bump pads and common bump pads, in which the common bump pad is commonly electrically connected to the first to third LED stacks, the first to third bump pads are electrically connected to the first to third LED stacks, respectively, and an upper surface of the n-electrode pad is located at substantially the same elevation as that of the lower p-electrode pad.

[0035] Hereinafter, the second LED stack is described as being disposed under the first LED stack, and the third LED stack is described as being disposed under the second LED stack, however, in some exemplary embodiments, the light emitting device may be flip-bonded. In this case, upper and lower positions of these first to third LED stacks may be reversed.

[0036] When the first to third LED stacks are stacked one above another, the light emitting device may have an increased luminous area of each sub-pixel without increasing a pixel area.

[0037] Furthermore, the upper surface of the n-electrode pad is set to be located at substantially the same elevation as that of the lower p-electrode pad, and thus, damage to any one of these pads may be prevented.

[0038] The first LED stack may emit light having a longer wavelength than that emitted from the second LED stack, and the second LED stack may emit light having a longer wavelength than that emitted from the third LED stack. For example, the first, second, and third LED stacks may emit red light, green light, and blue light, respectively. In another exemplary embodiment, the first LED stack may emit light having a longer wavelength than that emitted from the third LED stack, and the second LED stack may emit light having a shorter wavelength than that emitted from the third LED stack. For example, the first, second and third LED stacks may emit red light, blue light, and green light, respectively.

[0039] The first to third LED stacks may be independently driven, light generated from the first LED stack may be emitted to the outside through the second LED stack and the third LED stack, and light generated from the second LED stack may be emitted to the outside through the third LED stack.

[0040] Any one of the first to third transparent electrodes may be formed of a material different from the remaining transparent electrodes. For example, the first transparent electrode may be formed of indium-tin-oxide (ITO), and the second and third transparent electrodes may be formed of ZnO.

[0041] Each of the first to third transparent electrodes may contact the second conductivity type semiconductor layer, and the second and third transparent electrodes may be recessed to have a smaller area than that of the second conductivity type semiconductor layer of the second LED stack and that of the second conductivity type semiconductor layer of the third LED stack, respectively.

[0042] In this manner, as the second and third transparent electrodes are recessed, they may be

prevented from being damaged by etching gas during the manufacturing process.

[0043] Further, the common bump pad may be commonly electrically connected to first conductivity type semiconductor layers of the first to third LED stacks, in which the first to third bump pads may be electrically connected to second conductivity type semiconductor layers of first to third LED stacks, respectively. However, the inventive concepts are not limited thereto, the common bump pad may be commonly electrically connected to the second conductivity type semiconductor layers of the first to third LED stacks, and the first to third bump pads may be electrically connected to the first conductivity type semiconductor layers of the third LED stacks, respectively.

[0044] The light emitting device may further include an insulation layer covering side surfaces of the first to third LED stacks, in which the side surfaces of the first to third LED stacks and a side surface of the first transparent electrode may be in contact with the insulation layer, and side surfaces of the second and third transparent electrodes may be spaced apart from the insulation layer.

[0045] Side surfaces of the light emitting device may be inclined at a range of about 75 degrees to about 90 degrees with respect to the upper surface of the third LED stack. The inclination angle of the side surfaces of the light emitting device is set to about 75 degrees or more, and thus, a luminous area of the first LED stack may be secured.

[0046] The light emitting device may further include a first bonding layer interposed between the second LED stack and the third LED stack; and a second bonding layer interposed between the first LED stack and the second LED stack.

[0047] The light emitting device may further include lower through holes passing through the second LED stack and the first bonding layer to expose the n-electrode pad and the lower p-electrode pad, respectively; a lower common connector connected to the n-electrode pad; and a lower p-connector connected to the lower p-electrode pad, in which the lower common connector may be electrically connected to the first conductivity type semiconductor layer of the LED stack, and connected to the n-electrode pad exposed through the lower through hole, and the lower p-connector may be electrically connected to the lower p-electrode pad exposed through the lower through hole.

[0048] The light emitting device may further include an upper p-electrode pad which is disposed on the second transparent electrode and electrically connected to the second conductivity type semiconductor layer of the second LED stack.

[0049] The light emitting device may further include a through hole passing through the first LED stack to expose the first transparent electrode; through holes passing through the first LED stack, the first transparent electrode and the second bonding layer to expose the upper p-electrode pad, the lower p-connector, and the lower common connector, respectively; and first to third upper connectors and an upper common connector disposed on the first LED stack, and electrically connected to the first transparent electrode, the upper p-electrode pad, the lower p-connector, and the lower common connector through through holes passing through the first LED stack, in which the bump pads may be disposed on the first to third upper connectors and the upper common connector, respectively.

[0050] The bump pads may be located on flat portions of the first to third upper connectors and the upper common connector, respectively.

[0051] The light emitting device may further include an upper insulation layer covering the first to third upper connectors and the upper common connector, in which the upper insulation layer may have openings exposing the first to third upper connectors and the upper common connector, and each of the bump pads may be disposed in the openings.

[0052] The light emitting device may further include an intermediate insulation layer disposed between the first LED stack and the upper connectors, in which the intermediate insulation layer may cover a side surface of the light emitting device and a sidewall of the through hole passing

through the first LED stack, and may include openings exposing the first transparent electrode, the upper p-electrode pad, the lower p-connector, and the lower common connector.

[0053] The first to third LED stacks may be stacks separated from a growth substrate. As such, the light emitting device does not have the growth substrate.

[0054] A light emitting device for a display according to another exemplary embodiment includes: a first LED stack; a second LED stack disposed under the first LED stack; a third LED stack disposed under the second LED stack; a first transparent electrode interposed between the first LED stack and the second LED stack, and being in ohmic contact with a lower surface of the first LED stack; a second transparent electrode interposed between the first LED stack and the second LED stack, and being in ohmic contact with an upper surface of the second LED stack; a third transparent electrode interposed between the second LED stack and the third LED stack, and being in ohmic contact with an upper surface of the third LED stack; bump pads disposed on the first LED stack, in which each of the first to third LED stacks may include a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer, and at least one of the first to third transparent electrodes may be recessed from an edge of the second conductivity type semiconductor layer of the first to third LED stacks.

[0055] The light emitting device may further include an n-electrode pad disposed on the first conductivity type semiconductor layer of the third LED stack; a lower p-electrode pad disposed on the third transparent electrode, in which an upper surface of the n-electrode pad may be located at substantially the same elevation as that of the lower p-electrode pad.

[0056] A display apparatus according to an exemplary embodiment includes: a circuit board; and a plurality of light emitting devices arranged on the circuit board, in which each of the light emitting devices may be any one of the light emitting device set forth above, and the bump pads are electrically connected to the circuit board.

[0057] Hereinafter, exemplary embodiments of the inventive concepts will be described in detail with reference to the accompanying drawings.

[0058] FIG. 1 shows schematic perspective views illustrating display apparatuses according to exemplary embodiments.

[0059] The display apparatus according to exemplary embodiments may be used in a VR display apparatus, such as a smart watch **1000a** or a VR headset **1000b**, or an AR display apparatus, such as augmented reality glasses **1000c**. However, the inventive concepts are not limited thereto. The display apparatus may include a display panel for implementing an image.

[0060] FIG. 2 is a schematic plan view illustrating the display panel according to an exemplary embodiment.

[0061] Referring to FIG. 2, the display panel includes a circuit board **101** and light emitting devices **100**.

[0062] The circuit board **101** may include a circuit for passive matrix driving or active matrix driving. In an exemplary embodiment, the circuit board **101** may include interconnection lines and resistors. In another exemplary embodiment, the circuit board **101** may include interconnection lines, transistors, and capacitors. The circuit board **101** may also have pads disposed on an upper surface thereof to allow electrical connection to the circuit therein.

[0063] A plurality of light emitting devices **100** are arranged on the circuit board **101**. Each of the light emitting devices **100** may form one pixel. The light emitting device **100** includes bump pads **73**, and the bump pads **73** are electrically connected to the circuit board **101**. For example, the bump pads **73** may be bonded to pads exposed on the circuit board **101**.

[0064] An interval between the light emitting devices **100** may be greater than at least a width of the light emitting device **100**.

[0065] A configuration of the light emitting device **100** according to an exemplary embodiment will be described with reference to FIG. 3A, FIG. 3B, FIG. 3C, and FIG. 3D. FIG. 3A is a schematic plan view illustrating the light emitting device **100** according to an exemplary

embodiment. FIG. 3B, FIG. 3C, and 3D are schematic cross-sectional views taken along the lines A-A', B-B', and C-C' of FIG. 3A, respectively. Hereinafter, although the bump pads **73r**, **73b**, **73g**, and **73c** are exemplarily illustrated and described as being disposed at an upper side in FIG. 3A to FIG. 3D, the inventive concepts are not limited thereto. For example, in some exemplary embodiments, the light emitting device **100** may be flip-bonded on the circuit board **101** shown in FIG. 2, and in this case, the bump pads **73r**, **73b**, **73g**, and **73c** may be disposed at a lower side. [0066] Referring to FIG. 3A, FIG. 3B, FIG. 3C and FIG. 3D, the light emitting device **100** may include a first LED stack **23**, a second LED stack **33**, a third LED stack **43**, a first transparent electrode **25**, a second transparent electrode **35**, a third transparent electrode **45**, an n-electrode pad **47a**, a lower p-electrode pad **47b**, an upper p-electrode pad **53g**, a lower p-connector **53b**, a lower common connector **53c**, an upper common connector **63c**, a first upper connector **63r**, a second upper connector **63g**, a third upper connector **63b**, a first bonding layer **49**, a second bonding layer **59**, a lower insulation layer **51**, an intermediate insulation layer **61**, an upper insulation layer **71**, and bump pads **73a**, **73b**, **73c**, and **73d**. The light emitting device **100** may further include through holes **23h1**, **23h2**, **23h3**, and **23h4** passing through the first LED stack **23**, and through holes **33h1** and **33h2** passing through the second LED stack **33**.

[0067] As shown in FIG. 3B, the first, second, and third LED stacks **23**, **33**, and **43** according to an exemplary embodiment are stacked in the vertical direction. The first, second, and third LED stacks **23**, **33**, and **43** may be grown on different growth substrates from each other, and according to the illustrated exemplary embodiment, each of the growth substrates may be removed from the final light emitting device **100**. As such, the light emitting device **100** does not include the growth substrates of the first, second, and third LED stacks **23**, **33**, and **43**. However, the inventive concepts are not limited thereto, and in some exemplary embodiments, at least one of the growth substrates may be included in the light emitting device **100**.

[0068] Each of the first LED stack **23**, the second LED stack **33** and the third LED stack **43** includes a first conductivity type semiconductor layer **23a**, **33a**, or **43a**, a second conductivity type semiconductor layer **23b**, **33b**, or **43b**, and an active layer interposed therebetween. The active layer may have a multiple quantum well structure, for example.

[0069] The second LED stack **33** is disposed under the first LED stack **23**, and the third LED stack **43** is disposed under the second LED stack **33**. Light generated in the first to third LED stacks **23**, **33**, and **43** may be emitted to the outside through the third LED stack **43**.

[0070] In an exemplary embodiment, the first LED stack **23** may emit light having a longer wavelength than those emitted from the second and third LED stacks **33** and **43**, and the second LED stack **33** may emit light having a longer wavelength than that emitted from the third LED stack **43**. For example, the first LED stack **23** may be an inorganic light emitting diode emitting red light, the second LED stack **33** may be an inorganic light emitting diode emitting green light, and the third LED stack **43** may be an inorganic light emitting diode to emitting blue light. For example, the first LED stack **23** may include an AlGaInP-based well layer, the second LED stack **33** may include an AlGaInP or AlGaInN-based well layer, and the third LED stack **43** may include an AlGaInN-based well layer.

[0071] Since the first LED stack **23** emits light having a longer wavelength than those emitted from the second and third LED stacks **33** and **43**, light generated from the first LED stack **23** may be emitted to the outside after passing through the second and third LED stacks **33** and **43**. In addition, since the second LED stack **33** emits light having a longer wavelength than that of emitted from the third LED stack **43**, light generated from the second LED stack **33** may be emitted to the outside after passing through the third LED stack **43**.

[0072] In another exemplary embodiment, the first LED stack **23** may emit light having a longer wavelength than those emitted from the second and third LED stacks **33** and **43**, and the second LED stack **33** may emit light having a shorter wavelength than that emitted from the third LED stack **43**. For example, the first LED stack **23** may be an inorganic light emitting diode emitting red

light, the second LED stack **33** may be an inorganic light emitting diode emitting blue light, and the third LED stack **43** may be an inorganic light emitting diode emitting green light. For example, the first LED stack **23** may include an AlGaInP-based well layer, the second LED stack **33** may include an AlGaInN-based well layer, and the third LED stack **43** may include an AlGaInP-based or AlGaInN-based well layer.

[0073] A portion of light generated in the second LED **33** stack may be absorbed in the third LED stack **43**, and thus, luminous intensity of light emitted from the second LED stack **33** may be relatively lower than that of light emitted from the first or third LED stacks **23** or **43**. Accordingly, a ratio of luminance intensity of light emitted from the first to third LED stacks **23**, **33**, and **43** may be controlled.

[0074] According to the illustrated exemplary embodiment, the first conductivity type semiconductor layer **23a**, **33a** or **43a** of each of the LED stacks **23**, **33**, and **43** may be an n-type semiconductor layer, and the second conductivity type semiconductor layer **23b**, **33b** or **43b** thereof may be a p-type semiconductor layer. In addition, according to the illustrated exemplary embodiment, an upper surface of the first LED stack **23** may be an n-type semiconductor layer **23a**, an upper surface of the second LED stack **33** may be a p-type semiconductor layer **33b**, and an upper surface of the third LED stack **43** may be a p-type semiconductor layer **43b**. More particular, the first LED stack **23** has a stacked sequence of semiconductor layers different from those of the second and third LED stacks **33** and **43**. The semiconductor layers of the second LED stack **33** are stacked in the same order as the semiconductor layers of the third LED stack **43**, and thus, process stability may be enhanced, which will be described in more detail later with reference to a manufacturing method.

[0075] The second LED stack **33** includes a mesa etching region, in which a portion of the second conductivity type semiconductor layer **33b** is removed to expose an upper surface of the first conductivity type semiconductor layer **33a**. The third LED stack **43** also includes a mesa etching region, in which a portion of the second conductivity type semiconductor layer **43b** is removed to expose an upper surface of the first conductivity type semiconductor layer **43a**. The first LED stack **23**, however, may not include a mesa etching region. The through holes **33h1** and **33h2** may be formed in the mesa etching regions, and thus, sidewalls of the through holes **33h1** and **33h2** may have a stepped structure. In this case, since the first LED stack **23** does not include the mesa etching region, the through holes **23h1**, **23h2**, **23h3**, and **23h4** may have uniformly inclined sidewalls without having stepped sidewalls.

[0076] The third LED stack **43** according to an exemplary embodiment may have a flat lower surface, without being limited thereto. For example, the third LED stack **43** may include irregularities on a surface of the first conductivity type semiconductor layer **43a**, and light extraction efficiency may be improved by the irregularities. The irregularities on the surface of the first conductivity type semiconductor layer **43a** may be formed by separating a patterned sapphire substrate or by texturing the surface after separating the growth substrate, for example. In some exemplary embodiments, the first conductivity type semiconductor layer **33a** of the second LED stack **33** may also have a textured surface.

[0077] The first LED stack **23**, the second LED stack **33** and the third LED stack **43** according to the illustrated exemplary embodiment may be stacked to overlap one another, and may also have substantially the same luminous area. However, the luminous area of the first LED stack **23** may be smaller than that of the second LED stack **33**, and the luminous area of the second LED stack **33** may be smaller than that of the third LED stack **43**, by the through holes **23h1**, **23h2**, **23h3**, and **23h4** and the through holes **33h1** and **33h2**. In addition, a side surface of the light emitting device **100** may be inclined, such that a width of the light emitting device **100** may be gradually increasing from the first LED stack **23** to the third LED stack **43**. As such, the luminous area of the third LED stack **43** may be larger than that of the first LED stack **23**. An inclination angle of the side surface of the light emitting device **100** with respect to the upper surface of the third LED stack **43** may be

about 75 degrees to about 90 degrees. When the inclination angle is less than 75 degrees, the luminous area of the first LED stack **23** may become too small, and thus, it may be difficult to reduce a size of the light emitting device **100**.

[0078] The first transparent electrode **25** is disposed between the first LED stack **23** and the second LED stack **33**. The first transparent electrode **25** is in ohmic contact with the second conductivity type semiconductor layer **23b** of the first LED stack **23**, and transmits light generated by the first LED stack **23**. The first transparent electrode **25** may be formed using a transparent oxide layer or a metal layer, such as indium tin oxide (ITO). The first transparent electrode **25** may cover an entire surface of the second conductivity type semiconductor layer **23b** of the first LED stack **23**, and a side surface thereof may be disposed in parallel with a side surface of the first LED stack **23**. More particular, the side surface of the first transparent electrode **25** may not be covered with the second bonding layer **59**. Furthermore, the through holes **23h2**, **23h3**, and **23h4** may pass through the first transparent electrode **25**, and thus, the first transparent electrode **25** may be exposed by the sidewalls of the through holes **23h2**, **23h3**, and **23h4**. Meanwhile, the through hole **23h1** exposes an upper surface of the first transparent electrode **25**. However, the inventive concepts are not limited thereto, and in some exemplary embodiments, the first transparent electrode **25** may be partially removed along an edge of the first LED stack **23**, and thus, at least a portion of the side surface of the first transparent electrode **25** may be covered with the second bonding layer **59**. In addition, when the first transparent electrode **25** is previously patterned and removed in a region where the through holes **23h2**, **23h3**, and **23h4** are formed according to other exemplary embodiments, the first transparent electrode **25** may not be exposed by the sidewalls of the through holes **23h2**, **23h3**, and **23h4**.

[0079] A second transparent electrode **35** is in ohmic contact with the second conductivity type semiconductor layer **33b** of the second LED stack **33**. As shown in the drawings, the second transparent electrode **35** contacts the upper surface of the second LED stack **33** between the first LED stack **23** and the second LED stack **33**. The second transparent electrode **35** may be formed of a metal layer or a conductive oxide layer that is transparent to red light. For example, the conductive oxide layer may include SnO.sub.2, InO.sub.2, ITO, ZnO, IZO, or the like. In particular, the second transparent electrode **35** may be formed of ZnO, which may be formed as a single crystal on the second LED stack **33**. In this manner, the ZnO may have favorable electrical and optical characteristics as compared with the metal layer or other conductive oxide layers. In particular, ZnO has a strong bonding force to the second LED stack **33**, and remains undamaged even when the growth substrate is separated using a laser lift-off process during manufacture.

[0080] The second transparent electrode **35** may be partially removed along an edge of the second LED stack **33**, and, accordingly, an outer side surface of the second transparent electrode **35** is not exposed to the outside, but is covered with the lower insulation layer **51**. In particular, the side surface of the second transparent electrode **35** is recessed inwardly than that of the second LED stack **33**, and a region where the second transparent electrode **35** is recessed is filled with the lower insulation layer **51** and the second bonding layer **59**. The second transparent electrode **35** is also recessed near the mesa etching region of the second LED stack **33**, and the recessed region is filled with the lower insulation layer **51** and the second bonding layer **59**.

[0081] The third transparent electrode **45** is in ohmic contact with the second conductivity type semiconductor layer **43b** of the third LED stack **43**. The third transparent electrode **45** may be disposed between the second LED stack **33** and the third LED stack **43**, and contacts the upper surface of the third LED stack **43**. The third transparent electrode **45** may be formed of a metal layer or a conductive oxide layer that is transparent to red light and green light. For example, the conductive oxide layer may include SnO.sub.2, InO.sub.2, ITO, ZnO, IZO, or the like. In particular, the third transparent electrode **45** may be formed of ZnO, which may be formed as a single crystal on the third LED stack **43**. In this manner, the ZnO may have favorable electrical and optical characteristics as compared with the metal layer or other conductive oxide layers. In

particular, ZnO has a strong bonding force to the third LED stack **43**, and remains undamaged even when the growth substrate is separated using the laser lift-off process during manufacture.

[0082] The third transparent electrode **45** may be partially removed along an edge of the third LED stack **43**, and, accordingly, an outer side surface of the third transparent electrode **45** is not exposed to the outside, but is covered with the first bonding layer **49**. In particular, the side surface of the third transparent electrode **45** is recessed inwardly than that of the third LED stack **43**, and a region where the third transparent electrode **45** is recessed is filled with the first bonding layer **49**. The third transparent electrode **45** is also recessed near the mesa etching region of the third LED stack **43**, and the recessed region is filled with the first bonding layer **49**.

[0083] The second transparent electrode **35** and the third transparent electrode **45** are recessed as described above, and thus, the side surfaces of the second transparent electrode **35** and the third transparent electrode **45** may be prevented from being exposed to an etching gas, thereby improving the production yield of the light emitting device **100**.

[0084] According to an exemplary embodiment, the second transparent electrode **35** and the third transparent electrode **45** may be formed of the same kind of conductive oxide layer, for example, ZnO, and the first transparent electrode **25** may be formed of a different kind of conductive oxide layer from the second and third transparent electrodes **35** and **45**, such as ITO. However, the inventive concepts are not limited thereto, and each of the first to third transparent electrodes **25**, **35**, and **45** may be of the same kind, or at least one may be of a different kind.

[0085] The first to third transparent electrodes **25**, **35**, and **45** may be formed using a technique, such as thermal deposition, sputtering, sol-gel, hydrothermal synthesis, or the like. In particular, a transparent electrode formed through a chemically thin-film forming method, such as hydrothermal synthesis method, may generate a porous thin-film. In this case, voids in the porous thin-film may improve the light extraction efficiency of the LED stack, and may further relieve stress.

[0086] The voids may be controlled to be distributed at locations to enhance the optical properties of the LED stack. According to an exemplary embodiment, the voids may be distributed generally close to a side of the second conductivity type semiconductor layer **22b**, **33b**, and **43b** at one half point of the transparent electrode. The transparent electrode formed through the hydrothermal synthesis method may have voids and also have crystallinity, and, in particular, may be formed of a single crystal.

[0087] According to another exemplary embodiment, the voids may be distributed relatively uniformly over a wide area. The transparent electrode including the voids has an improved light extraction efficiency compared to a transparent electrode without the voids. The transparent electrode may be, for example, a ZnO layer or a doped ZnO layer. The doped ZnO layer may include at least one of, for example, silver (Ag), indium (In), tin (Sn), zinc (Zn), cadmium (Cd), gallium (Ga), aluminum (Al), magnesium (Mg), titanium (Ti), molybdenum (Mo), nickel (Ni), copper (Cu), gold (Au), platinum (Pt), rhodium (Rh), iridium (Ir), ruthenium (Ru), and palladium (Pd), as a dopant.

[0088] In an exemplary embodiment, the ZnO layer may also include a ZnO seed layer and a ZnO bulk layer. The ZnO seed layer has a relatively continuous surface. In addition, the ZnO seed layer and the ZnO bulk layer form a single crystal structure. In an exemplary embodiment, the ZnO seed layer and the ZnO bulk layer do not exhibit any interface between the ZnO seed layer and the ZnO bulk layer. In an exemplary embodiment, the ZnO seed layer has a thickness of several hundred angstroms. The ZnO seed layer may have, for example, a thickness of 200 angstroms or less. The ZnO bulk layer may have a thickness of 1 μm or less. In an exemplary embodiment, the ZnO bulk layer has a thickness of 8000 angstroms or less.

[0089] The n-electrode pad **47a** is in ohmic contact with the first conductivity type semiconductor layer **43a** of the third LED stack **43**. The n-electrode pad **47a** may be disposed on the first conductivity type semiconductor layer **43a** exposed through the second conductivity type semiconductor layer **43b**, that is, in the mesa etching region. The n-electrode pad **47a** may be

formed of, for example, Cr/Au/Ti. An upper surface of the n-electrode pad **47a** may be placed higher than that of the second conductivity type semiconductor layer **43b**, and further, higher than that of the third transparent electrode **45**. For example, a thickness of the n-electrode pad **47a** may be about 2 μm or more. The n-electrode pad **47a** may have a shape of a truncated cone, but is not limited thereto. The n-electrode pad **47a** may have various shapes, such as a square pyramid, a cylindrical shape, or a cylindrical shape.

[0090] The lower p-electrode pad **47b** may include substantially the same material as the n-electrode pad **47a**. An upper surface of the lower p-electrode pad **47b** is located at the substantially same elevation as the n-electrode pad **47a**, and, accordingly, a thickness of the lower p-electrode pad **47b** may be less than that of the n-electrode pad **47a**. More particularly, the thickness of the lower p-electrode pad **47b** may be approximately equal to a thickness of a portion of the n-electrode pad **47a** protruding above the third transparent electrode **45**. For example, the thickness of the lower p-electrode pad **47b** may be about 1.2 μm or less. The upper surface of the lower p-electrode pad **47b** is located at substantially the same elevation as that of the n-electrode pad **47a**, and thus, the lower p-electrode pad **47b** and the n-electrode pad **47a** may be simultaneously exposed when the through holes **33h1** and **33h2** are formed. When the elevations of the n-electrode pad **47a** and the lower p-electrode pad **47b** are different, any one of the electrode pads may be damaged in the etching process. As such, the elevations of the n-electrode pad **47a** and the lower p-electrode pad **47b** are set to be approximately equal, and thus, it is possible to prevent any one of the electrode pads from being damaged during the etching process or the like.

[0091] The first bonding layer **49** couples the second LED stack **33** to the third LED stack **43**. The first bonding layer **49** may be disposed between the first conductivity type semiconductor layer **33a** and the third transparent electrode **45**. The first bonding layer **49** may partially contact the second conductivity type semiconductor layer **43b**, and may partially contact the first conductivity type semiconductor layer **43a** exposed by the mesa etching region. In addition, the first bonding layer **49** may cover the n-electrode pad **47a** and the lower p-electrode pad **47b**.

[0092] The first bonding layer **49** may be formed of a transparent organic material layer, or may be formed of a transparent inorganic material layer. For example, the organic material layer may include SU8, poly methylmethacrylate (PMMA), polyimide, parylene, benzocyclobutene (BCB), or the like, and the inorganic material layer may include Al.sub.2O.sub.3 , SiO.sub.2 , SiNx , or the like. In addition, the first bonding layer **49** may be formed of spin-on-glass (SOG).

[0093] The through hole **33h1** and the through hole **33h2** pass through the second LED stack **33** and the first bonding layer **49** to expose the n-electrode pad **47a** and the lower p-electrode pad **47b**, respectively. As described above, the through holes **33h1** and **33h2** may be formed in the mesa etching region, and thus, the through holes **33h1** and **33h2** may have stepped sidewalls.

[0094] The lower insulation layer **51** is formed on the second LED stack **33**, and covers the second transparent electrode **35**. The lower insulation layer **51** also covers the sidewalls of the through holes **33h1** and **33h2**. The lower insulation layer **51** may have openings **51a** exposing the n-electrode pad **47a**, the lower p-electrode pad **47b**, the first conductivity type semiconductor layer **33a**, and the second transparent electrode **35**. The lower insulation layer **51** may be formed of a silicon oxide film or a silicon nitride film, and may be formed to have a thickness of, for example, about 800 nm.

[0095] The lower common connector **53c** may be disposed on the lower insulation layer **51**, and connected to the first conductivity type semiconductor layer **33a** and the n-electrode pad **47a** exposed through the openings **51a** of the lower insulation layer **51**. In particular, the lower common connector **53c** is connected to the first conductivity type semiconductor layer **33a** in the mesa etching region of the second LED stack **33**, and is connected to the n-electrode pad **47a** through the through hole **33h1**.

[0096] The lower p-connector **53b** may be disposed on the lower insulation layer **51**, and connected to the lower p-electrode pad **47b** exposed through the opening **51a** of the lower insulation layer **51**.

At least a portion of the lower p-connector **53b** is disposed on the lower insulation layer **51**.

[0097] The upper p-electrode pad **53g** may be disposed on the second transparent electrode **35** in the opening **51a** of the lower insulation layer **51**. As shown in FIG. 3A and FIG. 3C, the upper p-electrode pad **53g** may be disposed in the opening **51a** with a narrower width than that of the opening **51a**. However, the inventive concepts are not limited thereto, and in some exemplary embodiments, the width of the upper p-electrode pad **53g** may be greater than that of the opening **51a**, and a portion of the upper p-electrode pad **53g** may be disposed on the lower insulation layer **51**.

[0098] The lower common connector **53c**, the lower p-connector **53b**, and the upper p-electrode pad **53g** may be formed together in the same process and may include substantially the same material. For example, the lower common connector **53c**, the lower p-connector **53b**, and the upper p-electrode pad **53g** may include Ni/Au/Ti, and may be formed to have a thickness of about 2 μm .

[0099] The second bonding layer **59** couples the first LED stack **23** to the second LED stack **33**. As shown, the second bonding layer **59** may be disposed between the first transparent electrode **25** and the lower insulation layer **51**. The second bonding layer **59** may also cover the lower common connector **53c**, the lower p-connector **53b**, and the upper p-electrode pad **53g**. The second bonding layer **59** may also partially contact the second transparent electrode **35** exposed through the opening **51a** of the lower insulation layer **51**. The second bonding layer **59** may include substantially the same material that may form the first bonding layer **49** described above, and thus, repeated descriptions thereof will be omitted to avoid redundancy.

[0100] The through holes **23h1**, **23h2**, **23h3**, and **23h4** pass through the first LED stack **23**. The through hole **23h1** is formed to provide a passage for allowing electrical connection to the first transparent electrode **25**. In the illustrated exemplary embodiment, the through hole **23h1** exposes the upper surface of the first transparent electrode **25**, and does not pass through the first transparent electrode **25**. However, the inventive concepts are not limited thereto, and in some exemplary embodiments, the through hole **23h1** may pass through the first transparent electrode **25**, as long as the through hole **23h1** provides the passage for electrical connection to the first transparent electrode **25**.

[0101] The through holes **23h2**, **23h3**, and **23h4** may pass through the first LED stack **23**, and may also pass through the second bonding layer **59**. The through hole **23h2** exposes the upper p-electrode pad **53g**, the through hole **23h3** exposes the lower p-connector **53b**, and the through hole **23h4** exposes the lower common connector **53c**.

[0102] The through holes **23h1**, **23h2**, **23h3**, and **23h4** may be formed by etching the first conductivity type semiconductor layer **23a** and the second conductivity type semiconductor layer **23b** in the same process, and thus, the sidewalls of the through holes **23h1**, **23h2**, **23h3**, and **23h4** may have inclined surfaces without having a stepped structure.

[0103] The intermediate insulation layer **61** covers the first LED stack **23**, and covers the sidewalls of the through holes **23h1**, **23h2**, **23h3**, and **23h4**. The intermediate insulation layer **61** may also cover side surfaces of the first to third LED stacks **23**, **33**, and **43**. The intermediate insulation layer **61** may be patterned to have openings **61a** exposing a bottom portion of each of the through holes **23h1**, **23h2**, **23h3**, and **23h4**. The first transparent electrode **25**, the upper p-electrode pad **53g**, the lower p-connector **53b**, and the lower common connector **53c** may be exposed in the through holes **23h1**, **23h2**, **23h3**, and **23h4** by the openings **61a**. Further, the intermediate insulation layer **61** may have an opening **61b** exposing the upper surface of the first LED stack **23**, that is, the first conductivity type semiconductor layer **23a**. The intermediate insulation layer **61** may be formed of an aluminum oxide film, a silicon oxide film, or a silicon nitride film, and may be formed to have a thickness of, for example, about 800 nm.

[0104] The first upper connector **63r**, the second upper connector **63g**, the third upper connector **63b**, and the upper common connector **63c** are disposed on the intermediate insulation layer **61**. Each of the upper connectors **63r**, **63g**, **63b**, and **63c** is connected to the first transparent electrode

25, the upper p-electrode pad **53g**, and the lower p-connector **53b** exposed through the openings **61a** of the intermediate insulation layer **61**, respectively. Furthermore, the upper common connector **63c** may be connected to the first conductivity type semiconductor layer **23a** exposed to the opening **61b**.

[0105] The first upper connector **63r**, the second upper connector **63g**, the third upper connector **63b**, and the upper common connector **63c** may be formed of substantially the same material, for example, AuGe/Ni/Au/Ti, in the same process. In this case, AuGe may be in ohmic contact with the first conductivity type semiconductor layer **23a**. AuGe may be formed to have a thickness of about 100 nm, and Ni/Au/Ti may be formed to have a thickness of about 2 μm . In some exemplary embodiments, AuTe may replace AuGe.

[0106] The upper insulation layer **71** covers the intermediate insulation layer **61**, and covers the first upper connector **63r**, the second upper connector **63g**, the third upper connector **63b**, and the upper common connector **63c**. The upper insulation layer **71** may also cover the intermediate insulation layer **61** on the side surfaces of the first to third LED stacks **23**, **33**, and **43**. The upper insulation layer **71** may have openings **71a** exposing the first upper connector **63r**, the second upper connector **63g**, the third upper connector **63b**, and the upper common connector **63c**. The openings **71a** of the upper insulation layer **71** may be generally disposed on flat surfaces of the first upper connector **63r**, the second upper connector **63g**, the third upper connector **63b**, and the upper common connector **63c**. The upper insulation layer **71** may be formed of a silicon oxide film or a silicon nitride film, and may be formed thinner than the intermediate insulation layer **61**, for example, about 400 nm thick.

[0107] Each of the bump pads **73r**, **73g**, **73b**, and **73c** may be disposed on the first upper connector **63r**, the second upper connector **63g**, and the third upper connector **63b**, and the common connector **63c** in the openings **71a** of the upper insulation layer **71** and electrically connected thereto.

[0108] The first bump pad **73r** may be electrically connected to the second conductivity type semiconductor layer **23b** of the first LED stack **23** through the first upper connector **63r** and the first transparent electrode **25**.

[0109] The second bump pad **73g** may be electrically connected to the second conductivity type semiconductor layer **33b** of the second LED stack **33** through the second upper connector **63g**, the upper p-electrode pad **53g**, and the second transparent electrode **35**.

[0110] The third bump pad **73b** may be electrically connected to the second conductivity type semiconductor layer **43b** of the third LED stack **43** through the third upper connector **63b**, the lower p-connector **53b**, the lower p-electrode pad **47b**, and the third transparent electrode **45**.

[0111] The common bump pad **73c** may be electrically connected to the first conductivity type semiconductor layer **23a** of the first LED stack **23** through the upper common connector **63c**, electrically connected to the first conductivity type semiconductor layer **33a** of the second LED stack **33** through the lower common connector **53c**, and electrically connected to the first conductivity type semiconductor layer **43a** of the third LED stack **43** through the n-electrode pad **47a**.

[0112] As such, each of the first to third bump pads **73r**, **73g**, and **73b** may be electrically connected to the second conductivity type semiconductor layers **23b**, **33b**, and **43b** of the first to third LED stacks **23**, **33**, and **43**, and the common bump pad **73c** may be commonly electrically connected to the first conductivity type semiconductor layers **23a**, **33a**, and **43a** of the first to third LED stacks **23**, **33**, and **43**.

[0113] The bump pads **73r**, **73g**, **73b**, and **73c** may be disposed in the openings **71a** of the upper insulation layer **71**, and upper surfaces of the bump pads **73r**, **73g**, **73b**, and **73c** may be substantially flat. The bump pads **73r**, **73g**, **73b**, and **73c** may be disposed on the flat surfaces of the first to third upper connectors **63r**, **63g**, and **63b**, and the upper common connector **63c**. The bump pads **73r**, **73g**, **73b**, and **73c** may be formed of Au/In. For example, Au may be formed to have a

thickness of about 3 μm , and In may be formed to have a thickness of about 1 μm . According to an exemplary embodiment, the light emitting device **100** may be bonded to the pads on the circuit board **101** using In. However, the inventive concepts are not limited thereto, and in some exemplary embodiments, the light emitting device **100** may be bonded to the pads using Pb or AuSn of the bump pads.

[0114] In the illustrated exemplary embodiment, the upper surfaces of the bump pads **73r**, **73g**, **73b**, and **73c** are described and illustrated as being flat, but the inventive concepts are not limited thereto. For example, in some exemplary embodiments, the bump pads **73r**, **73g**, **73b**, and **73c** may have irregular upper surfaces, and some of the bump pads may be disposed on the upper insulation layer **71**.

[0115] According to the illustrated exemplary embodiment, the first LED stack **23** is electrically connected to the bump pads **73r** and **73c**, the second LED stack **33** is electrically connected to the bump pads **73g** and **73c**, and the third LED stack **43** is electrically connected to the bump pads **73b** and **73c**. Accordingly, cathodes of the first LED stack **23**, the second LED stack **33**, and the third LED stack **43** are electrically connected to the common bump pad **73c**, and anodes thereof are electrically connected to the first to third bump pads **73a**, **73b**, and **73c**, respectively. Accordingly, the first to third LED stacks **23**, **33**, and **43** may be driven independently.

[0116] A structure of the light emitting device **100** will be further described through a method of manufacturing the light emitting device **100** described below. FIG. 4A, FIG. 4B, and FIG. 4C are schematic cross-sectional views illustrating the first to third LED stacks grown on growth substrates, respectively, according to an exemplary embodiment.

[0117] First, referring to FIG. 4A, a first LED stack **23** including a first conductivity type semiconductor layer **23a** and a second conductivity type semiconductor layer **23b** is grown on a first substrate **21**. An active layer may be interposed between the first conductivity type semiconductor layer **23a** and the second conductivity type semiconductor layer **23b**.

[0118] The first substrate **21** may be a substrate capable of growing the first LED stack **23** thereon, such as a GaAs substrate. The first conductivity type semiconductor layer **23a** and the second conductivity type semiconductor layer **23b** may be formed of an AlGaInAs-based or AlGaInP-based semiconductor layer, and the active layer may include, for example, an AlGaInP-based well layer. A composition ratio of AlGaInP may be determined so that the first LED stack **23** emits red light, for example.

[0119] A first transparent electrode **25** may be formed on the second conductivity type semiconductor layer **23b**. As described above, the first transparent electrode **25** may be formed of a metal layer or a conductive oxide layer that transmits light generated by the first LED stack **23**, for example, red light. The first transparent electrode **25** may be formed of, for example, indium-tin oxide (ITO).

[0120] Referring to FIG. 4B, a second LED stack **33** including a first conductivity type semiconductor layer **33a** and a second conductivity type semiconductor layer **33b** is grown on a second substrate **31**. An active layer may be interposed between the first conductivity type semiconductor layer **33a** and the second conductivity type semiconductor layer **33b**.

[0121] The second substrate **31** may be a substrate capable of growing the second LED stack **33** thereon, such as a sapphire substrate, a GaN substrate, or a GaAs substrate. The first conductivity type semiconductor layer **33a** and the second conductivity type semiconductor layer **33b** may be formed of an AlGaInAs-based or AlGaInP-based semiconductor layer, an AlGaInN-based semiconductor layer, and the active layer may include, for example, an AlGaInP-based well layer or AlGaInN-based well layer. A composition ratio of AlGaInP or AlGaInN may be determined so that the second LED stack **33** emits green light, for example.

[0122] A second transparent electrode **35** may be formed on the second conductivity type semiconductor layer **33b**. As described above, the second transparent electrode **35** may be formed of a metal layer or a conductive oxide layer that transmits light generated by the first LED stack **23**,

for example, red light. In particular, the second transparent electrode **35** may be formed of ZnO.

[0123] Referring to FIG. **4C**, a third LED stack **43** including a first conductivity type semiconductor layer **43a** and a second conductivity type semiconductor layer **43b** is grown on a third substrate **41**. An active layer may be interposed between the first conductivity type semiconductor layer **43a** and the second conductivity type semiconductor layer **43b**.

[0124] The third substrate **41** may be a substrate capable of growing the third LED stack **43** thereon, such as a sapphire substrate, a SiC substrate, or a GaN substrate. In an exemplary embodiment, the third substrate **41** may be a flat sapphire substrate, but may also be a patterned sapphire substrate. The first conductivity type semiconductor layer **43a** and the second conductivity type semiconductor layer **43b** may be formed of an AlGaInN-based semiconductor layer, and the active layer may include, for example, an AlGaInN-based well layer. A composition ratio of AlGaInN may be determined so that the third LED stack **43** emits blue light, for example.

[0125] A third transparent electrode **45** may be formed on the second conductivity type semiconductor layer **43b**. As described above, the third transparent electrode **45** may be formed of a metal layer or a conductive oxide layer that transmits light generated in the first and second LED stacks **23** and **33**, for example, red light and green light. In particular, the third transparent electrode **45** may be formed of ZnO.

[0126] The first to third LED stacks **23**, **33**, and **43** are grown on the different growth substrates **21**, **31**, and **41**, respectively, and, accordingly, the order of the manufacturing process is not particularly limited.

[0127] Hereinafter, a method of manufacturing the light emitting device **100** using first to third LED stacks **23**, **33**, and **43** grown on growth substrates **21**, **31**, and **41** will be described.

Hereinafter, although a region of a single light emitting device **100** will be mainly illustrated and described, a plurality of light emitting devices **100** may be manufactured in a batch in the same manufacturing process using the LED stacks **23**, **33**, and **43** grown on the growth substrates **21**, **31**, and **41**.

[0128] FIGS. **5A**, **5B**, **5C**, **5D**, **6A**, **6B**, **6C**, **6D**, **7A**, **7B**, **7C**, **7D**, **8A**, **8B**, **8C**, **8D**, **9A**, **9B**, **9C**, **9D**, **10A**, **10B**, **10C**, **10D**, **11A**, **11B**, **11C**, **11D**, **12A**, **12B**, **12C**, **12D**, **13A**, **13B**, **13C**, and **13D** are schematic plan views and cross-sectional views illustrating the method of manufacturing the light emitting device **100** for a display according to an exemplary embodiment of the present disclosure. Hereinafter, the cross-sectional views are shown to correspond to those shown in FIGS. **3B**, **3C**, and **3D**, respectively.

[0129] First, referring to FIG. **5A**, FIG. **5B**, FIG. **5C**, and FIG. **5D**, the third transparent electrode **45** and the second conductivity type semiconductor layer **43b** of the third LED stack **43** are patterned to expose the first conductivity type semiconductor layer **43a** using photo and etching techniques. This process corresponds to, for example, a mesa etching process. A photoresist pattern may be used as an etching mask. For example, after the etching mask is formed, the third transparent electrode **45** may be etched first by a wet etching technique, and then the second conductivity type semiconductor layer **43b** may be etched by a dry etching technique using the same etching mask. In this manner, the third transparent electrode **45** may be recessed from a mesa etching region. FIG. **5A** exemplarily shows an edge of the mesa and does not show an edge of the third transparent electrode **45** to simplify illustrating. However, since the third transparent electrode **45** is wet etched using the same etching mask, the edge of the third transparent electrode **45** may also be recessed from the edge of the mesa toward an inner side of the mesa. Since the same etching mask is used, the number of photo processes may not be increased, thereby reducing the process costs. However, the inventive concepts are not limited thereto, and the etching mask for etching the mesa etching process may be different from the etching mask for etching the third transparent electrode **45**.

[0130] Subsequently, an n-electrode pad **47a** and a lower p-electrode pad **47b** are formed on the first conductivity type semiconductor layer **43a** and the third transparent electrode **45**, respectively.

The n-electrode pad **47a** and the lower p-electrode pad **47b** may be formed to have different thicknesses. In particular, an upper surface of the n-electrode pad **47a** and that of the lower p-electrode pad **47b** may be located at substantially the same elevation.

[0131] Referring to FIG. **6A**, FIG. **6B**, FIG. **6C**, and FIG. **6D**, the second LED stack **33** shown in FIG. **4B** is bonded onto the third LED stack **43** described with reference to FIG. **5A**, FIG. **5B**, FIG. **5C**, and FIG. **5D**. The second LED stack **33** is bonded to a temporary substrate using a temporary bonding/debonding (TBDB) technique, and the second substrate **31** is removed from the second LED stack **33**. The second substrate **31** may be removed using, for example, a laser lift off technique. After the second substrate **31** is removed, a roughened surface may be formed on a surface of the first conductivity type semiconductor layer **33a**. Thereafter, the first conductivity type semiconductor layer **33a** of the second LED stack **33** bonded to the temporary substrate may be disposed to face the third LED stack **43** and bonded to the third LED stack **43**. The second LED stack **33** and the third LED stack **43** are bonded to each other by a first bonding layer **49**. After bonding the second LED stack **33** to the third LED stack **43**, the temporary substrate may be removed using the laser lift off technique. Accordingly, the second LED stack **33** may be disposed on the third LED stack **43**, in which the second transparent electrode **35** may form an upper surface.

[0132] In general, when the second transparent electrode **35** is formed of ITO, ITO may be peeled from the second LED stack **33** when the second substrate **31** is removed using the laser lift off technique. As such, when the second substrate **31** is to be removed using the laser lift-off technique, the second transparent electrode **35** may include ZnO, which has a favorable bonding force.

[0133] Subsequently, the second transparent electrode **35** and the second conductivity type semiconductor layer **33b** are patterned to expose the first conductivity type semiconductor layer **33a**. The second transparent electrode **35** and the second conductivity type semiconductor layer **33b** may be patterned by using photo and etching techniques. This process may be performed using the wet etching and the dry etching techniques in substantially the same manner as the mesa etching process, during which the third transparent electrode **45** and the second conductivity type semiconductor layer **43b** are etched as described above.

[0134] For example, after the etching mask is formed, the second transparent electrode **35** may be etched first by the wet etching technique, and then the second conductivity type semiconductor layer **33b** may be etched by the dry etching technique using the same etching mask. Accordingly, the second transparent electrode **35** may be recessed from the mesa etching region. FIG. **6A** exemplarily shows an edge of the mesa, and does not show an edge of the second transparent electrode **35** to simplify illustration. However, since the second transparent electrode **35** is wet etched using the same etching mask, the edge of the second transparent electrode **35** may also be recessed from the edge of the mesa toward an inner side of the mesa. In this manner, since the same etching mask is used, the number of photo processes may not be increased, thereby reducing the process costs. However, the inventive concepts are not limited thereto, and in some exemplary embodiments, the etching mask for etching the mesa etching process and the etching mask for etching the second transparent electrode **35** may be different from each other.

[0135] As shown in FIG. **6A**, a mesa etching region of the second LED stack **33** may be partially overlapped with that of the third LED stack **43**. For example, a portion of the mesa etching region of the second LED stack **33** may be formed over the n-electrode pad **47a**. In addition, another portion of the mesa etching region thereof may be disposed over the lower p-electrode pad **47b**. In addition, a portion of the mesa etching region of the second LED stack **33** may be disposed on the mesa region of the third LED stack **43**.

[0136] Referring to FIG. **7A**, FIG. **7B**, FIG. **7C**, and FIG. **7D**, through holes **33h1** and **33h2** passing through the second LED stack **33** are formed. The through holes **33h1** and **33h2** pass through the first bonding layer **49** to expose the n-electrode pad **47a** and the lower p-electrode pad **47b**. The

through holes **33h1** and **33h2** may be formed in the mesa etching region, and thus, a stepped structure may be formed on sidewalls of the through holes **33h1** and **33h2**.

[0137] Since the upper surfaces of the lower p-electrode pad **47b** and the n-electrode pad **47a** are located at substantially the same elevation, any one of the pads may be prevented from being exposed and damaged during the formation of the through holes **33h1** and **33h2**.

[0138] Referring to FIG. **8A**, FIG. **8B**, FIG. **8C**, and FIG. **8D**, a lower insulation layer **51** is formed on the second LED stack **33**. The lower insulation layer **51** covers the second transparent electrode **35**, and covers the second conductivity type semiconductor layer **33b**. In addition, the lower insulation layer **51** covers the sidewalls of the through holes **33h1** and **33h2**. The lower insulation layer **51** may have openings **51a** exposing the second transparent electrode **35**, the first conductivity type semiconductor layer **33a**, the n-electrode pad **47a** and the lower p-electrode pad **47b**.

[0139] Subsequently, a lower common connector **53c**, a lower p-connector **53b**, and an upper p-electrode pad **53g** are formed on the lower insulation layer **51**. The lower common connector **53c**, the lower p-connector **53b**, and the upper p-electrode pad **53g** may be formed together with the same material.

[0140] An upper p-electrode pad **53g** may be disposed on the second transparent electrode **35** exposed in the opening **51a**. The lower p-connector **53b** is connected to the lower p-electrode pad **47b** exposed through the opening **51a**, and is also partially disposed on the lower insulation layer **51**. The lower common connector **53c** is connected to the first conductivity type semiconductor layer **33a** and the n-electrode pad **47a** exposed through the openings **51a**, and a portion of the lower common connector **53c** is disposed on the lower insulation layer **51**.

[0141] Referring to FIG. **9A**, FIG. **9B**, FIG. **9C** and FIG. **9D**, the first LED stack **23** of FIG. **4A** is bonded to the second LED stack **33**. The first LED stack **23** and the second LED stack **33** may be bonded using a second bonding layer **59**, so that the first transparent electrode **25** faces the second LED stack **33**. Accordingly, the second bonding layer **59** is in contact with the first transparent electrode **25**, and also is in contact with the lower insulation layer **51**, the lower p-connector **53b**, the upper p-electrode pad **53g**, and the lower common connector **53c**, and, further, in contact with the second transparent electrode **35** exposed in a periphery of the lower p-connector **53b**. The first substrate **21** is removed from the first LED stack **23**. The first substrate **21** may be removed using, for example, an etching technique.

[0142] Referring to FIG. **10A**, FIG. **10B**, FIG. **10C**, and FIG. **10D**, through holes **23h1**, **23h2**, **23h3**, and **23h4** passing through the first LED stack **23** are formed. The through hole **23h1** exposes the first transparent electrode **25**, and the through holes **23h2**, **23h3**, and **23h4** pass through the second bonding layer **59** to expose the upper p-electrode pad **53g**, the lower p-connector **53b**, and the lower common connector **53c**, respectively. Since the through holes **23h1** and the through holes **23h2**, **23h3**, and **23h4** have different depths, they may be formed by different processes. Meanwhile, the through holes **23h2**, **23h3**, and **23h4** may be formed together in the same process because the depths thereof are substantially the same.

[0143] The through holes **23h1**, **23h2**, **23h3**, and **23h4** may be formed to pass through the first LED stack **23**, and thus, sidewalls of the through holes may be formed without steps, unlike those of the through holes **33h1** and **33h2**.

[0144] Referring to FIG. **11A**, FIG. **11B**, FIG. **11C**, and FIG. **11D**, an isolation trench is formed to define a region of the light emitting device **100** by an isolation process. The isolation trench may expose the third substrate **41** along peripheries of the first to third LED stacks **23**, **33**, and **43**. Between regions of the light emitting device, the isolation trench may be formed by sequentially removing the first LED stack **23**, the first transparent electrode **25**, the second bonding layer **59**, the lower insulation layer **51**, the second LED stack **33**, the first bonding layer **49**, and the third LED stack **43**. The second transparent electrode **35** and the third transparent electrode **45** are not exposed during the isolation process, and thus, the second transparent electrode **35** and the third transparent

electrode **45** may not be damaged by etching gas. When the second and third transparent electrodes **35** and **45** are formed of ZnO, ZnO may be easily damaged by etching gas. However, according to the illustrated exemplary embodiment, the second transparent electrode **35** and the third transparent electrode **45** may be prevented from being exposed to an etching gas by forming the second and third transparent electrodes **35** and **45** to be recessed inwardly.

[0145] In the illustrated exemplary embodiment, the first to third LED stacks **23**, **33**, and **43** are described as being sequentially patterned through the isolation process, but the inventive concepts are not limited thereto. For example, in some exemplary embodiments, the third LED stack **43** may be removed in advance in a region where the isolation trench will be formed before bonding the second LED stack **33**, or the second LED stack **33** may be removed in advance in the region in which the isolation trench will be formed before bonding the first LED stack **23**. In this case, the region where the third LED stack **43** is removed may be filled with the first bonding layer **49**, and the region where the second LED stack **33** is removed may be filled with the second bonding layer **59**. Accordingly, the second and third LED stacks **33** and **43** may not be exposed in the isolation process.

[0146] Referring to FIG. 12A, FIG. 12B, FIG. 12C, and FIG. 12D, an intermediate insulation layer **61** is formed on the first LED stack **23**. The intermediate insulation layer **61** may cover side surfaces of the first to third LED stacks **23**, **33**, and **43**, side surfaces of the first and second bonding layers **49** and **59**, a side surface of the transparent electrode **25**, and a side surface of the lower insulation layer **51** exposed through the isolation trench.

[0147] The intermediate insulation layer **61** may also cover sidewalls of the through holes **23h1**, **23h2**, **23h3**, and **23h4**. However, the intermediate insulation layer **61** is patterned to have openings **61a** exposing bottoms of the through holes **23h1**, **23h2**, **23h3**, and **23h4**, and an opening **61b** exposing the first conductivity type semiconductor layer **23a** of the first LED stack **23**. The openings **61a** expose the first transparent electrode **25**, the upper p-electrode pad **53g**, the lower p-connector **53b**, and the lower common connector **53c** in the through holes **23h1**, **23h2**, **23h3**, and **23h4**.

[0148] First to third upper connectors **63r**, **63g**, and **63b**, and an upper common connector **63c** are formed on the intermediate insulation layer **61**. The first upper connector **63r** is connected to the first transparent electrode **25**, the second upper connector **63g** is connected to the upper p-electrode pad **53g**, and the third upper connector **63b** is connected to the lower p-connector **53b**. The upper common connector **63c** may be connected to the lower common connector **53c**.

[0149] Referring to FIG. 13A, FIG. 13B, FIG. 13C, and FIG. 13D, an upper insulation layer **71** is formed to cover the intermediate insulation layer **61** and the connectors **63r**, **63g**, **63b**, and **63c**. The upper insulation layer **71** may also cover the intermediate insulation layer **61** on the side surfaces of the first to third LED stacks **23**, **33**, and **43**. However, the upper insulation layer **71** may be patterned to have openings **71a** exposing the first to third upper connectors **63r**, **63g**, and **63b**, and the upper common connector **63c**.

[0150] Subsequently, bump pads **73r**, **73g**, **73b**, and **73c** are formed in the openings **71a**, respectively. The first bump pad **73r** is disposed on the first upper connector **63r**, the second bump pad **73g** is disposed on the second upper connector **63g**, and the third bump pad **73b** is disposed on the third upper connector **63b**. The common bump pad **73c** is disposed on the upper common connector **63c**.

[0151] Then, the light emitting device **100** is bonded onto a circuit board **101** (see FIG. 14), and the third substrate **41** may be separated to from the light emitting device **100**. A schematic cross-sectional view of the light emitting device **100** bonded to the circuit board **101** is exemplarily shown in FIG. 14.

[0152] Although FIG. 14 exemplarily illustrates a single light emitting device **100** disposed on the circuit board **101**, however, a plurality of light emitting devices **100** may be mounted on the circuit board **101**. Each of the light emitting devices **100** may form one pixel capable of emitting any one

of blue light, green light, and red light, and a plurality of pixels are arranged on the circuit board **101** to provide a display panel.

[0153] The plurality of light emitting devices **100** may be formed on the substrate **41**, and the light emitting devices **100** may be transferred onto the circuit board **101** in a group, not individually. FIG. **15A**, FIG. **15B**, and FIG. **15C** are schematic cross-sectional views illustrating a method of transferring the light emitting device to the circuit board according to an exemplary embodiment. Hereinafter, a method of transferring the light emitting devices **100** formed on the substrate **41** to the circuit board **101** in a group will be described.

[0154] Referring to FIG. **15A**, as described above, when the process of FIG. **13A**, FIG. **13B**, of FIG. **13C**, and FIG. **13D** are completed, the plurality of light emitting devices **100** are isolated from each other, and are arranged on the substrate **41** by the isolation trench.

[0155] Meanwhile, the circuit board **101** having pads on an upper surface thereof is provided. The pads are arranged on the circuit board **101** to correspond to locations where the pixels for a display are to be arranged. In general, an interval between the light emitting devices **100** arranged on the substrate **41** may be more dense than that of the pixels on the circuit board **101**.

[0156] Referring to FIG. **15B**, bump pads of the light emitting devices **100** are selectively bonded to the pads on the circuit board **101**. The bump pads and the pads may be bonded using In bonding, for example. In this case, the light emitting devices **100** located between pixel regions may be spaced apart from the circuit board **101**, since these light emitting devices **100** do not have pads of the circuit board **101** to be bonded to.

[0157] Subsequently, a mask **201** is disposed on the substrate **41** and is irradiated with a laser. A light transmitting region of the mask **201** is disposed to correspond to the light emitting devices **100** bonded to the circuit board **101**, and thus, the light emitting devices **100** bonded to the pads of the circuit board **101** are selectively irradiated with the laser. Thereafter, the light emitting devices **100** are transferred to the circuit board **101** by separating the light emitting devices **100** irradiated with the laser from the substrate **41**. Accordingly, the display panel in which the light emitting devices **100** are arranged on the circuit board **101** is provided. The display panel may be mounted on various display apparatuses as described with reference to FIG. **1**.

[0158] Although certain exemplary embodiments and implementations have been described herein, other embodiments and modifications will be apparent from this description. Accordingly, the inventive concepts are not limited to such embodiments, but rather to the broader scope of the appended claims and various obvious modifications and equivalent arrangements as would be apparent to a person of ordinary skill in the art.

Claims

1. A light emitting device for a display, comprising: a circuit board; a plurality of light emitting units arranged on the circuit board, each light emitting unit comprising a first LED stack including: a first conductivity type semiconductor layer; a second conductivity type semiconductor layer; a first electrode disposed on the first conductivity type semiconductor layer; and a second electrode disposed on the second conductivity type semiconductor layer; a plurality of bump pads disposed between the plurality of light emitting units and the circuit board; and a bonding layer disposed between the second electrode and the circuit board, wherein the second electrode has a side surface recessed inwardly with respect to a side surface of the light emitting unit to define a recessed portion, and the bonding layer is filled in the recessed portion.
